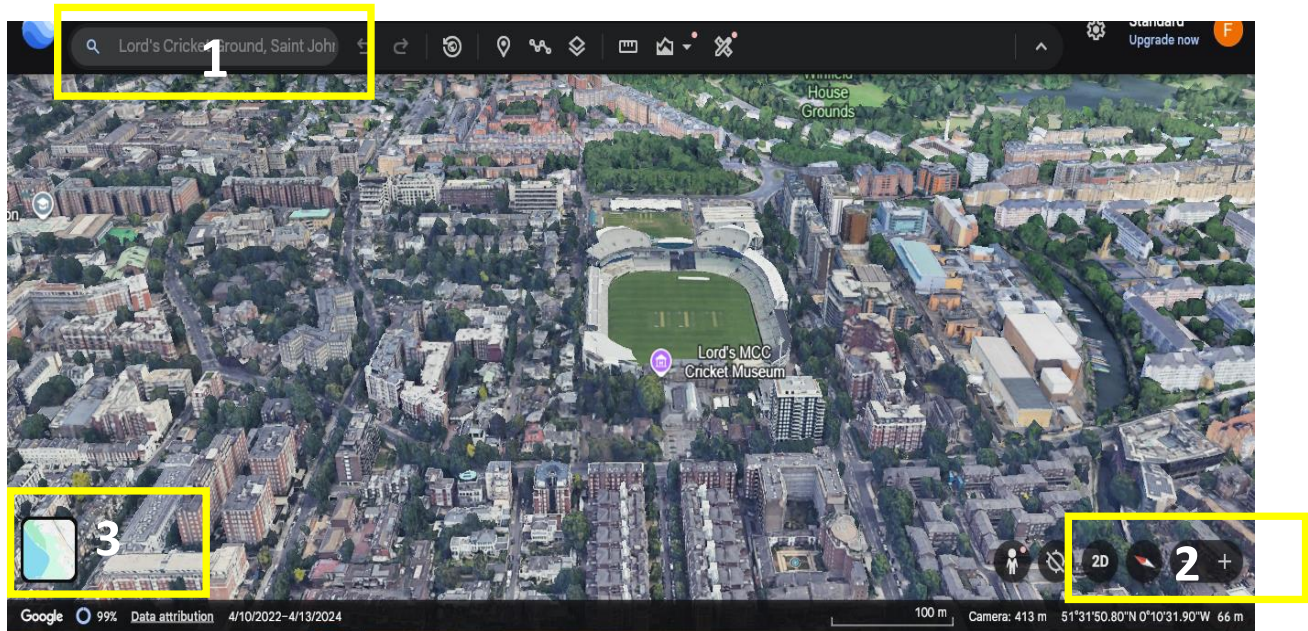


HCI Focus:



1. Search Bar & Query Input Field

Provides a direct text input mechanism with autocomplete capabilities, allowing the user to bypass manual map navigation and instantly jump to specific global coordinates or named landmarks. It reduces cognitive load by prioritizing goal-directed search over manual spatial exploration.

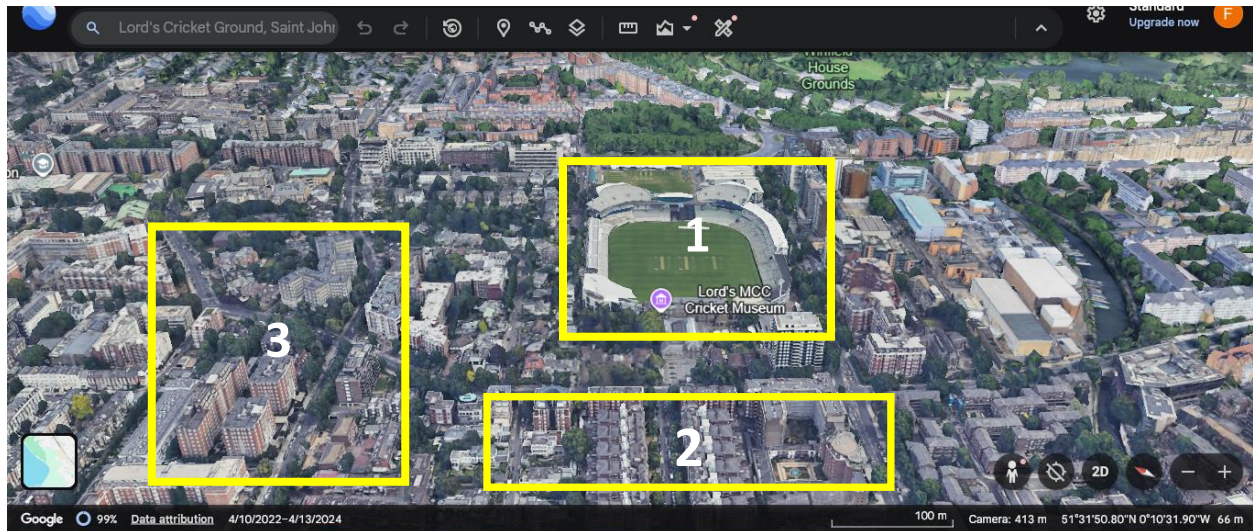
2. Navigation & View Control Toolbar

Groups interactive affordances—including the compass for orientation, the 2D/3D perspective toggle, and zoom buttons (+/-)—into a consolidated control dock. It gives the user immediate manual control to manipulate altitude, tilt, and perspective without needing keyboard shortcuts.

3. Basemap / Layer Style Switcher

Displays an interactive thumbnail preview that shows the current rendering layer (such as satellite imagery, clean map, or 3D terrain). It acts as an affordance that lets the user toggle visual styles and data overlays with a single click, providing immediate visual feedback of the active display state.

Computer Graphics Focus:



1. 3D Mesh Geometry (Stadium Architecture)

Represents the physical stadium structure using interconnected vertices, edges, and polygon faces. Instead of remaining a flat 2D satellite photograph, the graphics engine reconstructs volumetric boundaries, stepped grandstand tiers, and curved outer walls to give the landmark true 3D spatial presence.

2. Texture Mapping (Roof & Ground Surfaces)

High-resolution 2D photographic images are wrapped and UV-mapped onto the underlying polygonal meshes of the residential blocks and roadways. This projects photorealistic colors, brick patterns, roof shingles, and pavement details directly onto the 3D geometry without requiring millions of individual micro-polygons.

3. Perspective Projection & Occlusion Cues (Foreground Buildings)

Demonstrates a perspective camera view where parallel street lines foreshorten and converge toward the background. The elevated foreground apartment blocks partially block (occlude) the ground and trees behind them, providing powerful depth cues that help the human brain perceive relative distance and real-world scale in a virtual 3D environment.

Reflection Questions:

1. How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

High-quality 3D photogrammetric meshes and perspective camera depth cues significantly enhance spatial awareness, allowing users to accurately evaluate real-world building heights, topography, and landmarks. However, rendering complex 3D geometry and dense texture maps can introduce visual clutter and frame rate lag on lower-end devices, occasionally making camera panning and precise click selections feel sluggish.

2. What is one simple HCI change you would make to improve the user interface?

I would replace the low-contrast, static zoom and perspective controls with dynamic, floating on-canvas gesture handles and high-contrast tooltip indicators to make primary orientation controls immediately discoverable for touch and mouse users alike.